

A High-Dimensional Gaussian Counterexample to the Proposed Brownian Besov Comparison

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Abstract

Hytönen and Veraar proved that the expected critical Besov norm of an X -valued Brownian motion is universally comparable to the p -th moment of its endpoint. They asked whether their upper estimate by $\sqrt{p} \mathbb{E}\|W(1)\|$ admits a universal reverse estimate. We give a negative answer. For standard Brownian motion in ℓ_2^d , the two quantities have orders $\sqrt{d} + \sqrt{p}$ and $\sqrt{pd}$, respectively. Thus $d = p \rightarrow \infty$ is a counterexample family. More precisely, the correct universal formula is

$$\mathbb{E}\|W\|_{B_{p,\infty}^{1/2}(0,1;X)} \asymp \mathbb{E}\|W(1)\| + \sqrt{p} \sigma(W(1)),$$

where σ is the weak Gaussian standard deviation.

1 Source question

For $0 < \alpha < 1$ and $1 \leq p \leq \infty$, the source equips the vector-valued Besov space with the equivalent dyadic norm

$$\|f\|_{B_{p,\infty}^\alpha(0,1;X)} = \|f\|_{L^p(0,1;X)} + \sup_{n \geq 0} 2^{n\alpha} \|f(\cdot + 2^{-n}) - f\|_{L^p(0,1-2^{-n};X)}.$$

The source explicitly declares that an unsubscripted comparison sign has a universal constant. Its Theorem 6.1 and the unresolved sentence in Remark 6.1 are reproduced below from page 15 of the arXiv PDF [1].

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Theorem 6.1. *Let X be a Banach space. Let $p \in [1, \infty)$. For an X -valued Brownian motion W we have*

$$(6.3) \quad \mathbb{E}\|W\|_{B_{p,\infty}^{1/2}(0,1;X)} \approx (\mathbb{E}\|W(1)\|^p)^{1/p},$$

$$(6.4) \quad \mathbb{E}\|W\|_{B_{\Phi_2,\infty}^{1/2}(0,1;X)} \approx \mathbb{E}\|W(1)\|.$$

Remark 6.1. By [9, Corollary 3.2], the estimate (6.3) implies that

$$\mathbb{E}\|W\|_{B_{p,\infty}^{1/2}(0,1;X)} \lesssim \sqrt{p} \mathbb{E}\|W(1)\|,$$

but we do not know if there is a two sided comparison here. The above estimate is also an immediate consequence of (6.4) and the definition of the various norms.

Thus the precise question is whether there exists a universal $c > 0$ such that for every real Banach space X , every X -valued Brownian motion W , and every $p \geq 1$,

$$\mathbb{E}\|W\|_{B_{p,\infty}^{1/2}(0,1;X)} \geq c\sqrt{p}\mathbb{E}\|W(1)\|. \quad (1)$$

For a centered X -valued Gaussian random variable G , write

$$m(G) := \mathbb{E}\|G\|, \quad \sigma(G) := \sup_{\|x^*\| \leq 1} (\mathbb{E}|x^*(G)|^2)^{1/2}.$$

2 Proof intuition

The source has already reduced the Besov expectation to the p -th moment of $G = W(1)$. A Gaussian p -th moment has two independent scales: its mean $m(G)$ and its fluctuation scale $\sqrt{p}\sigma(G)$. The questioned bound replaces the weak scale $\sigma(G)$ by the potentially much larger strong scale $m(G)$. High-dimensional Euclidean Gaussians maximize this mismatch: $m(G) \asymp \sqrt{d}$ while $\sigma(G) = 1$. Choosing $p = d$ makes the proposed right-hand side larger than the true scale by a factor of order $\sqrt{d}$.

3 The corrected estimate

Lemma 1 (Strong and weak Gaussian scales). *There are universal constants $c, C > 0$ such that every centered Gaussian random variable G in a real Banach space and every $p \geq 1$ satisfy*

$$c(m(G) + \sqrt{p}\sigma(G)) \leq (\mathbb{E}\|G\|^p)^{1/p} \leq C(m(G) + \sqrt{p}\sigma(G)).$$

Proof. The upper estimate is the standard Gaussian concentration moment bound: the Gaussian isoperimetric inequality gives a subgaussian upper tail above $m(G)$ with variance proxy $\sigma(G)^2$, and integration of that tail gives

$$(\mathbb{E}\|G\|^p)^{1/p} \leq m(G) + C\sqrt{p}\sigma(G).$$

See, for example, Ledoux–Talagrand [2, Chapter 3].

For the converse, monotonicity of moments gives $(\mathbb{E}\|G\|^p)^{1/p} \geq m(G)$. Also, for every $x^* \in X^*$ with $\|x^*\| \leq 1$,

$$(\mathbb{E}\|G\|^p)^{1/p} \geq (\mathbb{E}|x^*(G)|^p)^{1/p} = \gamma_p(\mathbb{E}|x^*(G)|^2)^{1/2},$$

where $\gamma_p = (\mathbb{E}|N(0,1)|^p)^{1/p} \asymp \sqrt{p}$. Taking the supremum over x^* yields the lower bound by $c\sqrt{p}\sigma(G)$. The maximum of the two lower bounds is at least half their sum. $\square$

Theorem 1 (Correct universal Besov comparison). *There are universal constants $c, C > 0$ such that, for every real Banach space X , every X -valued Brownian motion W , and every $p \geq 1$,*

$$c(\mathbb{E}\|W(1)\| + \sqrt{p}\sigma(W(1))) \leq \mathbb{E}\|W\|_{B_{p,\infty}^{1/2}(0,1;X)} \leq C(\mathbb{E}\|W(1)\| + \sqrt{p}\sigma(W(1))).$$

Proof. Theorem 6.1 of Hytönen–Veraar [1] states, with universal constants,

$$\mathbb{E}\|W\|_{B_{p,\infty}^{1/2}(0,1;X)} \asymp (\mathbb{E}\|W(1)\|^p)^{1/p}.$$

Apply Lemma 1 to the centered Gaussian endpoint $W(1)$. $\square$

4 The counterexample

Theorem 2 (Negative answer to Remark 6.1). *The universal reverse estimate (1) is false. In fact, for standard d -dimensional Brownian motion W_d with values in $X_d = \ell_2^d$,*

$$\mathbb{E}\|W_d\|_{B_{p,\infty}^{1/2}(0,1;\ell_2^d)} \asymp \sqrt{d} + \sqrt{p}, \quad \sqrt{p} \mathbb{E}\|W_d(1)\|_2 \asymp \sqrt{pd}.$$

Consequently, the ratio of the left-hand side of (1) to its right-hand scale tends to zero along $d = p \rightarrow \infty$.

Proof. Let $g_d = W_d(1) = (g_1, \dots, g_d)$, where the coordinates are independent standard normal random variables. The dual of ℓ_2^d is ℓ_2^d , so

$$\sigma(g_d) = \sup_{\|u\|_2 \leq 1} (\mathbb{E}|\langle u, g_d \rangle|^2)^{1/2} = 1.$$

Moreover, $m(g_d) \asymp \sqrt{d}$. The upper bound follows from Jensen's inequality. For an elementary lower bound, put $Y = \|g_d\|_2^2$. Then $\mathbb{E}Y = d$ and $\mathbb{E}Y^2 = d^2 + 2d \leq 3d^2$; Paley–Zygmund gives $\mathbb{P}(Y \geq d/2) \geq 1/12$, hence $\mathbb{E}\sqrt{Y} \geq \sqrt{d/2}/12$.

Theorem 1 now yields the first displayed comparison, while the estimate for $m(g_d)$ yields the second. If (1) held with a universal $c > 0$, setting $d = p = n$ would force a quantity bounded above by a universal multiple of $2\sqrt{n}$ to be bounded below by a positive universal multiple of n . This is impossible as $n \rightarrow \infty$. $\square$

5 Verification, limitations, and novelty

The included script evaluates the exact chi-moment identity

$$\mathbb{E}\|g_d\|_2^p = 2^{p/2} \frac{\Gamma((d+p)/2)}{\Gamma(d/2)}$$

and confirms numerically that $\|g_n\|_2 / (\sqrt{n} \mathbb{E}\|g_n\|_2)$ decreases toward zero. This is a sanity check, not part of the proof.

The result answers only the universal, dimension-free interpretation fixed by the source's notation. For a fixed nonzero Gaussian G , a reverse bound with a constant depending on $\sigma(G)/m(G)$ follows from Lemma 1.

A bounded search on 22 July 2026 covered the run's four cheap indexes, its 35,297-paper deterministic source-signal corpus, exact arXiv/web searches for the source title, authors, quoted unresolved phrase, and formula variants, and visible later primary papers citing the source. No explicit later answer to Remark 6.1 or this high-dimensional counterexample was found. Novelty confidence is moderate, not definitive.

Human-review recommendation. Verify that the source's Theorem 6.1 is uniform in X, p, W , and check the standard concentration upper bound in Lemma 1. The remaining Euclidean calculation is elementary.

References

- [1] T. P. Hytönen and M. C. Veraar, *On Besov regularity of Brownian motions in infinite dimensions*, Probability and Mathematical Statistics **28** (2008), no. 1, 143–162; [arXiv:0801.2959](#).
- [2] M. Ledoux and M. Talagrand, *Probability in Banach Spaces: Isoperimetry and Processes*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 23, Springer, 1991.